Multimodal Retrieval-Augmented Generation using Image-Text Embeddings

Saptarshi Mukherjee[†]

Abstract

This report presents a multimodal Retrieval-Augmented Generation (RAG) pipeline for image captioning. The system combines image-text embedding models for retrieval with vision-language models for generation. Given an input image, similar captions are retrieved using cosine similarity between CLIP-based embeddings and provided as context to a language model.

Experiments are conducted on Flickr30k and MS COCO(only validation set) datasets, comparing RAG against non-RAG setups, different language models (Phi-3.5 vs LLaVA-1.5), and different embedding encoders (ViT-B/16 vs ViT-B/32). Results show that RAG consistently improves performance across multiple metrics, with gains strongly dependent on retrieval quality. The study highlights that retrieval effectiveness is the primary bottleneck in multimodal RAG systems. This invites further study on effectiveness of rerankers and other sophisticated retrieval techniques.

1. Methodology

1.1. Pipeline

The system follows a retrieval-augmented pipeline:

- Load pre-computed caption embeddings(using CLIP model)into RAM.
- Take the input image from the user
- Compute image embeddings for the input image.
- Retrieve top-k captions using cosine similarity(can also use other techniques like FAISS for approximate search).
- Provide retrieved captions as context to a vision-language model.
- Generate the final caption by getting the answer from the LLM

A non-RAG baseline is also evaluated, where the retrieval step is omitted.

1.2. Retrieval Evaluation

Retrieval quality is evaluated using Hit@K metrics:

- Hit@K = 1 if any of the top-K retrieved captions belong to the same image.
- Hit@K = 0 otherwise.

We try out values of K like 1, 5 and 10 to study how changing K affects answer quality and how much increasing K can help in improving answer quality.

2. Datasets

2.1. Flickr30k

Flickr30k contains approximately 31,000 images with 158,000 captions, averaging five captions per image. The dataset provides diverse natural language descriptions, making it suitable for building robust caption embeddings.

2.2. MS COCO

The MS COCO val2017 subset (5,000 images) is used due to computational constraints. Each image includes approximately five captions, offering diverse real-world descriptions for retrieval-based learning.

3. Evaluation Metrics

We evaluate performance using:

- BLEU: n-gram precision (0–1) $\rightarrow$ higher = more exact wording overlap; strong gains show better caption accuracy.
- ROUGE-1: unigram recall (0–1) $\rightarrow$ higher = better coverage of key objects/concepts.
- ROUGE-2: bigram recall (0–1) $\rightarrow$ higher = better phrase-level coherence.
- ROUGE-L: sequence overlap (0–1) $\rightarrow$ higher = more fluent, well-structured sentences.

- CLIPScore: image–text semantic similarity (0–1 typical) → higher = better alignment with visual content.

4. Working of the pipeline for a demo image

Figure 1. Demo image from MSCOCO

```
Top retrieved captions:

0.321 | A black Honda motorcycle with a dark burgundy seat.
0.300 | A very nice looking motorcycle with some long handle bars.
0.296 |  A black motorcycle parked in a parking lot in front of a grass filled hill
0.291 | A black Honda motorcycle parked in front of a garage.
0.287 | Large black motorcycle sitting next to a white building.
```

Figure 2. Retrieved captions

We construct a prompt by combining the input image with retrieved captions to guide the vision-language model. The retrieved captions are first formatted into a context block and appended to the instruction.

Prompt format:

```
<|user|>
<|image_1|>
Describe this image in one sentence.

Helpful retrieved captions:
- caption_1
- caption_2
...
- caption_k

Only answer with the description sentence.
<|end|>
<|assistant|>
```

The retrieved captions provide additional context to the model, enabling more accurate and grounded generation.

4.1. Generation Procedure

The prompt and image are passed to the processor to obtain tokenized inputs and image embeddings. These are then fed into the vision-language model for generation.

- The model generates output tokens autoregressively.
- Greedy decoding is used (no sampling) for deterministic outputs.
- The maximum number of generated tokens is limited to 25.

Formally, the generated output is obtained as:

$$y = \text{Model}(x_{\text{image}}, x_{\text{prompt}}) \tag{1}$$

4.2. Post-processing

The generated text is post-processed to ensure clean and concise output:

- Remove special tokens (e.g., `<|user|>`, `<|assistant|>`).
- Truncate at predefined stop markers.
- Extract only the first sentence.

This ensures that the final output is a single, well-formed descriptive sentence-just like the ground truth captions, for a fair comparison.

5. Results

5.1. RAG vs No RAG comparison

	Metric	RAG	No-RAG	Gain
0	BLEU	0.2556 ± 0.2767	0.2058 ± 0.1635	0.049774
1	ROUGE-1	0.5617 ± 0.2216	0.5458 ± 0.1345	0.015949
2	ROUGE-2	0.3514 ± 0.2926	0.3032 ± 0.1482	0.048154
3	ROUGE-L	0.5306 ± 0.2355	0.4955 ± 0.1363	0.035076
4	CLIPScore	0.3269 ± 0.0353	0.3208 ± 0.0444	0.006076

Figure 3. RAG vs no RAG: Flickr dataset scores, standard deviation also shown

	Metric	RAG	No-RAG	Gain
0	BLEU	0.5385 ± 0.3942	0.2147 ± 0.2025	0.323844
1	ROUGE-1	0.7507 ± 0.2444	0.5821 ± 0.1379	0.168613
2	ROUGE-2	0.6121 ± 0.3494	0.3269 ± 0.1655	0.285224
3	ROUGE-L	0.7269 ± 0.2642	0.5281 ± 0.1487	0.198774
4	CLIPScore	0.3385 ± 0.0395	0.3242 ± 0.0475	0.014313

Figure 4. RAG vs no RAG: MSCOCO dataset scores, standard deviation also shown

RAG improves performance across all metrics, with larger gains on MS COCO, indicating the importance of richer caption diversity.

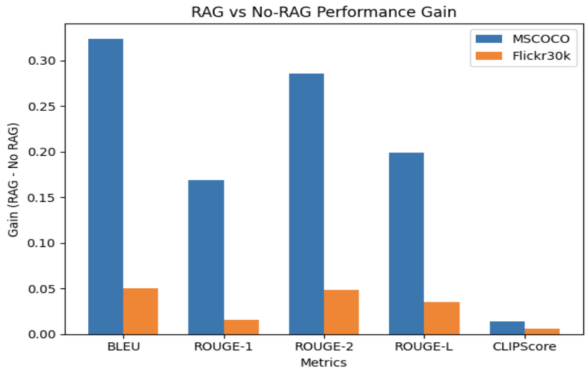

Figure 5. RAG vs no RAG: Gain

5.2. LLM comparison: Phi-3.5-vision-instruct vs Llava-1.5b-7b-hf

	Metric	Phi-3.5	LLaVA-1.5-7B	Gain (Phi-LLaVA)
0	BLEU	0.4227 ± 0.3479	0.3715 ± 0.3048	0.051261
1	ROUGE-1	0.6789 ± 0.2620	0.6588 ± 0.2194	0.020010
2	ROUGE-2	0.5212 ± 0.3648	0.4676 ± 0.3009	0.053613
3	ROUGE-L	0.6548 ± 0.2800	0.6275 ± 0.2338	0.027373
4	CLIPScore	0.3440 ± 0.0341	0.3359 ± 0.0382	0.008154

Figure 6. Phi vs Llava: Flickr dataset scores, standard deviation also shown

	Metric	Phi-3.5	LLaVA-1.5-7B	Gain (Phi-LLaVA)
0	BLEU	0.5090 ± 0.3893	0.4340 ± 0.3606	0.075029
1	ROUGE-1	0.7299 ± 0.2391	0.7070 ± 0.2180	0.022811
2	ROUGE-2	0.5773 ± 0.3450	0.5229 ± 0.3136	0.054479
3	ROUGE-L	0.7028 ± 0.2588	0.6747 ± 0.2358	0.028046
4	CLIPScore	0.3393 ± 0.0351	0.3313 ± 0.0385	0.008000

Figure 7. Phi vs Llava: MSCOCO dataset scores, standard deviation also shown

- Experiments have been done on 500 random samples from the respective datasets.
- Phi-3.5 outperforms LLaVA across all metrics, with statistically significant differences.
- Phi-3.5-vision-instruct is a more recent and optimized vision-language model with approximately 4.2B parameters, designed for strong multimodal reasoning and instruction following, which likely contributes to its superior performance.
- In contrast, LLaVA-1.5-7B, while larger at around 7B parameters, is based on an earlier architecture with limited alignment between visual and textual representations, resulting in weaker generation quality.

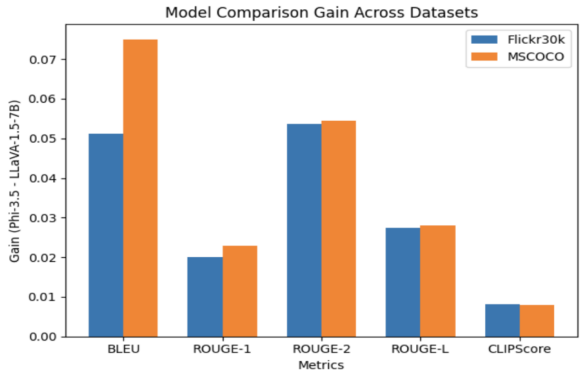

Figure 8. Phi3.5 vs Llava-1.5

	Metric	p-value	Significant (<0.05)
0	BLEU	4.134686e-06	True
1	ROUGE-1	2.245471e-02	True
2	ROUGE-2	1.582618e-04	True
3	ROUGE-L	8.761427e-03	True
4	CLIPScore	4.453916e-09	True

Figure 9. Significance test: Phi vs Llava

5.3. Encoder Comparison

	Metric	ViT-B/16	ViT-B/32	Gain (B16-B32)
0	BLEU	0.2473 ± 0.2748	0.2554 ± 0.2768	-0.008090
1	ROUGE-1	0.5651 ± 0.2181	0.5618 ± 0.2213	0.003278
2	ROUGE-2	0.3529 ± 0.2879	0.3508 ± 0.2926	0.002113
3	ROUGE-L	0.5351 ± 0.2308	0.5308 ± 0.2352	0.004210
4	CLIPScore	0.3143 ± 0.0424	0.3267 ± 0.0352	-0.012397

Figure 10. Flickr dataset:ViT-B/16 vs ViT-B/32 encoder

- Experiments have been done on 500 random samples from the respective datasets.
- While BLEU and ROUGE show no significant differences ($p > 0.05$), CLIPScore shows a statistically significant difference, indicating that encoder choice affects image-text alignment even if answer quality remains similar.

5.4. Recall Study

- Observe that not finding at least one ground truth caption in the list of retrieved captions really worsens the answer quality.

	Metric	p-value	Significant (<0.05)
0	BLEU	4.480745e-01	False
1	ROUGE-1	7.004362e-01	False
2	ROUGE-2	8.483155e-01	False
3	ROUGE-L	6.349323e-01	False
4	CLIPScore	2.645738e-12	True

Figure 11. Significance test on Flickr: ViT-B/16 vs ViT-B/32

	Metric	ViT-B/16	ViT-B/32	Gain (B16-B32)
0	BLEU	0.5097 ± 0.3955	0.5080 ± 0.3915	0.001629
1	ROUGE-1	0.7314 ± 0.2370	0.7281 ± 0.2354	0.003328
2	ROUGE-2	0.5707 ± 0.3501	0.5702 ± 0.3426	0.000580
3	ROUGE-L	0.7025 ± 0.2574	0.7021 ± 0.2537	0.000392
4	CLIPScore	0.3291 ± 0.0400	0.3359 ± 0.0353	-0.006795

Figure 12. MSCOCO dataset:ViT-B/16 vs ViT-B/32 encoder

- Interestingly, increasing K actually worsens the quality, although after accounting for the greater number of images for which a better answer is obtained, the difference is compensated.
- In general, providing irrelevant contextual information worsens answer quality, so there is a tradeoff between ensuring enough answers are good vs the quality of the answers where the correct captions were retrieved.

5.5. Link to the github repository

Link: GitHub Repository

Discussion and Future Work

Discussion and Conclusions

- This project demonstrates that multimodal Retrieval-Augmented Generation (RAG) significantly improves image captioning performance across multiple evaluation metrics. The improvements are consistent across both Flickr30k and MS COCO datasets, with larger gains observed on MS COCO, likely due to its more diverse and complex captions.
- Experimentally, we observe that stronger vision-language models such as Phi-3.5 yield better performance than earlier models like LLaVA, highlighting the importance of improved multimodal alignment.
- In contrast, the choice of embedding encoder (ViT-B/16 vs ViT-B/32) has minimal impact on surface-level caption quality, although it affects semantic alignment as measured by CLIPScore.

	Metric	p-value	Significant (<0.05)
0	BLEU	9.175858e-01	False
1	ROUGE-1	7.322247e-01	False
2	ROUGE-2	9.667620e-01	False
3	ROUGE-L	9.701482e-01	False
4	CLIPScore	5.508269e-08	True

Figure 13. Significance test on MSCOCO: ViT-B/16 vs ViT-B/32

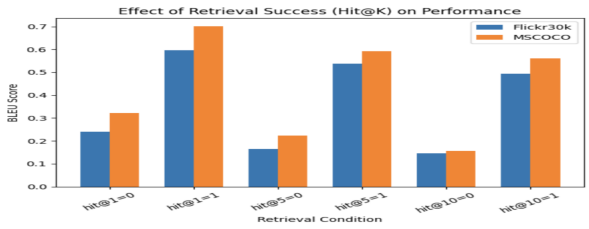

Figure 14. Recall study(Hit@1, Hit@5, Hit@10 for BLEU scores) for Flickr and MSCOCO

- Most importantly, the recall study reveals that retrieval quality is the primary factor influencing RAG performance. When relevant captions are successfully retrieved (Hit@K = 1), the generated outputs improve significantly, whereas irrelevant retrievals degrade performance. Additionally, increasing the number of retrieved captions introduces a trade-off between improved coverage and increased noise.
- Overall, the results suggest that while advances in model architecture contribute to performance gains, improving retrieval mechanisms remains the key direction for enhancing multimodal RAG systems.

Future Work

- Improve retrieval (reranking to boost Hit@K): Introduce a second-stage reranker on top of CLIP retrieval to better prioritize the most relevant captions, improving retrieval accuracy and downstream generation quality.
- Adaptive top-k instead of fixed k: Dynamically select the number of retrieved captions based on similarity scores or confidence thresholds, reducing noise from irrelevant captions.
- Fine-grained (entity/phrase-level) retrieval: Move beyond full-caption retrieval by retrieving specific phrases or entity-level descriptions, enabling more precise and targeted contextual grounding.
- Test stronger / different retrievers: Explore alternative embedding models or multimodal retrievers to improve alignment between image and text representations.

- Scale to larger datasets (full MS COCO): Extend experiments to larger datasets to improve robustness of results. This can be done after obtaining enough GPU compute.
- Testing larger models like Phi-3.5-vision variants and newer LLaVA-based architectures to study how further increase in model capacity affects multimodal reasoning and caption quality.
- Add human evaluation: Complement automatic metrics with human evaluation to assess relevance, fluency, and factual correctness of generated captions.